The diagonal Ramsey iteration of Gupta, Ndiaye, Norin and Wei, verified and continued: $R(k,k) \le (3.77213\ldots)^{k+o(k)}$ conditional on their Theorem 14, and a benchmark certificate refuted

Carlos Toledo*

October 2026 — draft v0.1 for review; every number interpolated from the record; machine-derived, not peer-reviewed; author list to be agreed

Abstract

Gupta, Ndiaye, Norin and Wei (arXiv 2407.19026v2) rewrite the Campos–Griffiths–Morris–Sahasrabudhe argument as one inductive statement, their Theorem 14: a bound $R(k,\ell) \le e^{F(\ell/k)k+o(k)}$ follows from four conditions on functions F, M, X, Y of $\lambda = \ell/k$, and their Lemma 15 turns each bound so obtained into a larger admissible region for the next round. Two rounds give their Theorem 1, $R(k,k) \le (3.7992\ldots)^{k+o(k)}$. After Remark 17 the paper prints a third round proposed by a language model as "preliminary, unverified", with base $3.78233\ldots$. We decide the numerical hypothesis of Theorem 14 for that round, with a continuous witness M chosen here and Y taken from Lemma 15 for the proved bound $F_{0.03}$: it holds on all of $(0,1]$, decided by two programs written apart in different languages and arithmetics, and the base is $3.7823287755373116\ldots$, of which the printed 3.78233 is its rounding. We then continue the iteration for 5 more rounds, each proposed by a float optimiser and each decided in the region the round before it establishes, reaching $R(k,k) \le (3.7721307629\ldots)^{k+o(k)}$, in particular $R(k,k) \le 3.7722^{\,k+o(k)}$, conditional on the paper's Theorem 14, Lemma 15 and Theorem 1; the two programs agree on 25 digits of the final base. In floating point the iteration converges, near 3.7732 at degree six and 3.77213 at degree nine, so further rounds of this kind buy almost nothing, which is consistent with the authors' expectation that going below 3.75 needs new ideas. None of these bases is the best known bound: $R(k,k) \le 3.769^{\,k+o(k)}$ (Neisler, Shin and Sukhatankar, kernel-checked in Lean by its authors) and $R(k,k) \le 3.69507^{\,k}$ (Lu and Wang) were obtained by other methods in September 2026 and are not checked here. Finally we re-decide a certificate that the HorizonMath benchmark (arXiv 2603.15617v2) credits to a language model and reports as verified, claiming $R(k,k) \le (3.6960\ldots)^{k+o(k)}$ in the same framework. Its pair $(X(1), Y(1)) = (0.22745,\ 0.9988)$, at the point that sets the constant, lies outside the region Theorem 14 requires: Erdős's 1947 lower bound excludes it. The benchmark's checker accepts a pair when either of two inequalities holds where membership needs both. What is refuted is the certificate, not the inequality.

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

1 Introduction

The Ramsey number $R(k,\ell)$ is the least N such that every red–blue colouring of the edges of K_N contains a red K_k or a blue K_ℓ [1]. Erdős and Szekeres proved $R(k,k) \le 4^k$ in 1935 [2]; Erdős's random colouring gives $R(k,k) > 2^{k/2}$ [3]; the polynomial savings of Thomason, Conlon and Sah [4, 5, 6] left the base 4 untouched until Campos, Griffiths, Morris and Sahasrabudhe proved $R(k,k) \le (4-\varepsilon)^k$ [7]. Gupta, Ndiaye, Norin and Wei [8] shorten that proof, replacing the book algorithm by an inductive statement, and optimise its parameters: their Theorem 1 gives $R(k,k) \le (3.7992\ldots)^{k+o(k)}$. Two lower bounds have appeared since, by other methods, and neither is checked here: $R(k,k) \le 3.769^{\,k+o(k)}$ by Neisler, Shin and Sukhatankar [10], a self-consistent bootstrap of the same book algorithm whose whole proof is kernel-checked in Lean 4 by its authors, and $R(k,k) \le 3.69507^{\,k}$ for all sufficiently large k by Lu and Wang [11]. What this paper decides is the iteration of [8] itself; none of its bases is the best known bound.

The optimisation has a fixed shape. Theorem 14 of [8] takes a function F and three auxiliary functions on $(0,1]$ and concludes $R(k,\ell) \le e^{F(\ell/k)k+o(k)}$ for $\ell \le k$, provided four conditions hold at every $\lambda \in (0,1]$; one of them places a pair $(X(\lambda), Y(\lambda))$ in a region R of pairs that known Ramsey bounds allow. Lemma 15 converts a bound already established into a curve of pairs inside R. So each round of Theorem 14 enlarges the region available to the next, and the proof of Theorem 1 is two rounds. The paper's own verification of the numerical hypotheses is interval arithmetic with directed rounding, certified by Mathematica (their Lemma 16 and Appendix A); its AI disclosure records that the numerical calculations in the earliest public version were incorrectly justified and corrected later, and that the main results were formalised in Lean 4. That history is the reason for the present paper: in this method the mathematics is the authors' and the risk sits in a one-variable inequality that must hold on a whole interval.

After Remark 17 the paper prints one more round, which we quote from the pinned PDF (page 20):

> We asked ChatGPT 5.6 Sol to perform an additional iteration of optimization in Theorem 14. A preliminary, unverified iteration suggests that Theorem 1 holds with
>
> $$G_{\mathrm{AI}}(\lambda) = e^{-\lambda}\big(-0.3864\lambda + 0.8347\lambda^2 - 2.0156\lambda^3 + 2.7171\lambda^4 - 1.7541\lambda^5 + 0.4522\lambda^6\big).$$
>
> If verified it would improve the upper bound on the diagonal Ramsey numbers to $R(k,k) \le (3.78233\ldots)^{k+o(k)}$. Further improvements by performing additional iterations are possible, but we expect that lowering the base of the exponent below 3.7 and, likely, even below 3.75 would require new ideas.

The remark names neither the M nor the Y that would go with G_{AI}. Theorem 14 asks only that some exist; a verification is therefore a search for a witness followed by a decision on all of $(0,1]$.

A second source of claims is a benchmark. HorizonMath [9] poses "improve the base c in $R(k,k) \le c^{k+o(k)}$" as a problem in the framework of [8], with an automatic checker, and reports that a language model produced a certificate with $c \approx 3.6960$, accepted by the checker, "verified by domain experts" and marked PASS (its Appendix A.3). That base is below the 3.75 the authors of [8] expect to need new ideas.

This paper does three things on those two sources, in one grammar. A claim is read from the pinned source; the numerical statement it rests on is *decided* by a program that returns a verdict together with a witness, and whose refusal is itself a recorded outcome; and every number printed here is interpolated from the resulting record, which is public and re-runnable. We say

a one-variable inequality is *certified* when an interval enclosure with every rounding outward decides it true on a finite cover of its domain; the word is used in no other sense. Results of other people are *re-decided*, never otherwise characterised. The arithmetic itself is standard [12, 13], and the authors of [8] use it too; what is new is the record: the witness is published with the verdict, the decision is made by two programs that share no code, and the refusal of forged inputs is part of the evidence.

2 Statement of results

Throughout, $h(\lambda) = (1+\lambda)\log(1+\lambda) - \lambda\log\lambda$, so that $\log\binom{k+\ell}{\ell} = h(\ell/k)k + o(k)$ uniformly for $1 \le \ell \le k$, and $F_b = h + g_b$ with $g_b(\lambda) = (-\frac{1}{4}\lambda + b\lambda^2 + 0.08\lambda^3)e^{-\lambda}$, as in [8]; Theorem 1 there is the bound $F_{0.03}$.

Theorem 1 (conditional)**.** *Assume Theorem 14, Lemma 15 and Theorem 1 of [8] (version 2407.19026v2). Let $q_1, \dots, q_6$ be the polynomials of Table 1 (q_1 of degree 6, the others of degree 9) and $F_i = h + q_i e^{-\lambda}$. Then for every $i \le 6$ and all positive integers $\ell \le k$,*

$$R(k,\ell) \le e^{F_i(\ell/k)k + o(k)},$$

and in particular

$$R(k,k) \le c^{\,k+o(k)} \le 3.7722^{\,k+o(k)},$$
$$c = e^{F_6(1)} \in [3.772130762900366389246867 1,\ 3.772130762900366389246867 2].$$

Theorem 2 (the remark's round)**.** *Under the same assumptions, Theorem 1 of [8] holds with $G = G_{\mathrm{AI}}$; that is, for $\ell \le k$,*

$$R(k,\ell) \le e^{G_{\mathrm{AI}}(\ell/k)k + o(k)}\binom{k+\ell}{\ell}, \qquad \text{and} \qquad R(k,k) \le (3.7823287755373116\dots)^{k+o(k)}.$$

The printed 3.78233 *is its rounding of this base, which is $4e^{G_{\mathrm{AI}}(1)}$ with $e\,G_{\mathrm{AI}}(1) = -0.1521$.*

Theorem 2 is round 1 of Theorem 1. The proofs are the decisions of Section 5: for each round, with the witness M_i of Section 4 and $Y_i = Y_{F_{i-1}}(X_i)$ from Lemma 15 (with $F_0 = F_{0.03}$), the four conditions of Theorem 14 hold at every $\lambda \in (0,1]$, decided on a finite cover of $(0,1]$ by intervals by two programs; and for every i the hypotheses of Lemma 15 for $f = F_{i-1}$ are decided first, so that the region of round i is one the round before it has established (for $i = 1$ this re-decides what Remark 17 of [8] asserts for $F_{0.03}$). What is *decided* is exactly this: the numerical hypotheses of two theorems of [8], for specific functions, on all of $(0,1]$. What is *assumed* is Theorem 14, Lemma 15 and Theorem 1 of [8], including the uniformity of their $o(k)$ terms, which the paper states and which the induction through Lemma 15 uses.

Proposition 1 (unconditional)**.** *Let (X,Y) be the pair of the HorizonMath certificate ([9], Appendix A.3) at $\lambda = 1$: $X(1) = 0.22745427\dots$, $Y(1) = 0.9988$. Then $(X(1),Y(1)) \notin R$. Consequently the certificate does not satisfy the hypotheses of Theorem 14 of [8], and it proves nothing about $R(k,k)$. Of its 201 decided points, 56 lie outside R and 128 more are not placed in R by the bound the problem names.*

The proposition is proved in Section 6 from Erdős's bound [3] and the definition of R; it depends on no unproved statement. It refutes a certificate. Whether $R(k,k) \le (3.6960\dots)^{k+o(k)}$ is true is not decided here, in either direction.

3 Theorem 14 and Lemma 15, as used

We restate what is used, in the paper's notation. Let R_0 be the set of pairs $(x,y) \in (0,1)^2$ for which there is N with

$$R(k,\ell) \le x^{-k}y^{-\ell} \qquad \text{for all } k,\ell \in \mathbb{N} \text{ with } k+\ell \ge N, \tag{1}$$

let R be its closure and R^* its interior ([8], (12)). Observation 10 there records that $(x,1-x) \in R$ for $0 \le x \le 1$ (the Erdős–Szekeres bound), that R is closed downward in each coordinate, and that $R(k,\ell) \le x^{-k-o(k)}y^{-\ell-o(\ell)}$ for a pair in $(0,1)^2$ already puts the pair in R. Since $R(k,\ell) = R(\ell,k)$ and (1) quantifies over all k,ℓ, the set R is symmetric; this is the point on which Section 6 turns.

Theorem 3 ([8], Theorem 14)**.** *Let $F : (0,1] \to \mathbb{R}_+$ be smooth and let $M, X, Y : (0,1] \to (0,1)$ with M continuous be such that*

$$F'(\lambda) > 0, \qquad X(\lambda) = \big(1-e^{-F'(\lambda)}\big)^{\frac{1}{1-M(\lambda)}}(1-M(\lambda)), \qquad (X(\lambda),Y(\lambda)) \in R,$$
$$\text{and} \qquad F(\lambda) > -\tfrac{1}{2}\big(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)\big)$$

for all $0 < \lambda \le 1$. Then $R(k,\ell) \le e^{F(\ell/k)k+o(k)}$ for all $k \ge \ell$.

Write $\mathrm{slack}(\lambda) = F(\lambda) + \frac{1}{2}(\log X + \lambda \log M + \lambda \log Y)$ for the margin of the last condition. Two of the hypotheses cost nothing: the F of every round below is $h + qe^{-\lambda}$ with q a polynomial, real-analytic on $(0,1]$, and M is continuous by construction; and positivity of F follows from the strict inequality, since each logarithm is of a number in $(0,1)$ and so negative. $X \in (0,1)$ follows from $F' > 0$ and $M \in (0,1)$. What remains to decide at every λ is $F' > 0$, $M \in (0,1)$, $(X,Y) \in R$ and slack > 0.

Lemma 1 ([8], Lemma 15)**.** *Let $f : (0,1] \to \mathbb{R}$ be differentiable and strictly concave with $f' > 0$, and suppose $R(k,\ell) \le \exp(f(\ell/k)k + o(k))$ for $1 \le \ell \le k$. Put $A(t) = e^{-f'(t)}$ and $B(t) = e^{tf'(t)-f(t)}$, and suppose $A < B$ on $(0,1]$, that A increases from 0 to $a = A(1)$ and B decreases from 1 to $b = B(1)$. Then*

$$Y_f(x) = \begin{cases} B(t), & 0 < x < a,\ A(t) = x, \\ e^{-f(1)}/x, & a \le x \le b, \\ A(t), & b < x < 1,\ B(t) = x, \end{cases}$$

is well defined, takes values in $(0,1)$, and $(x, Y_f(x)) \in R$ for $0 < x < 1$.

Its proof places both $(A(t),B(t))$ and $(B(t),A(t))$ in R for every t, by applying the hypothesis once with $\ell \le k$ and once, through $R(k,\ell) = R(\ell,k)$, with $\ell > k$: both orientations are proved before a point is placed. Remark 17 of [8] applies the lemma to $f = F_{0.03}$, the bound of Theorem 1, so $(x, Y_{F_{0.03}}(x)) \in R$ for every $x \in (0,1)$; that curve is the Y of round 1. For round i the lemma is applied to $f = F_{i-1}$, the bound round $i-1$ establishes ($F_0 = F_{0.03}$). Its hypotheses for such an $f = h + qe^{-t}$ are decided, for every round's f including $F_{0.03}$, as follows: $f'' < 0$ and $f' > 0$ on $[10^{-9}, 1]$ by intervals, and below 10^{-9} because $-1/(t(1+t))$ and $\log(1/t)$ dominate a polynomial part whose size is bounded in terms of the coefficients; $A' = -f''A > 0$ and $B' = tf''B < 0$, with $A \to 0$ and $B \to 1$ as $t \downarrow 0$ because $f' \to \infty$, $f(0^+) = 0$ and $q(0) = 0$; and $A < B$ because $\log(B/A) = (1+t)f'(t) - f(t)$ decreases and equals $2f'(1) - f(1)$ at $t = 1$, which is decided positive. Appendix A of [8] notes that on the three branches $(\log Y_f)' = -\kappa(\log X)'$ with $\kappa = 1/t$, 1 and t; the first program below uses the same fact.

4 The witness and the two programs

The witness M. For each round $M(\lambda) = \lambda\, m(\lambda)$ with m continuous and piecewise linear through 101 rational nodes at $\lambda = j/100$, $j = 0, \dots, 100$, each printed with six decimals. At the interior nodes the value is the float maximiser of the slack; at $\lambda = 0$ it is the root μ of $1/\mu + 1/(E+\mu) = 1$ with $E = e^{-q'(0)}$, the choice that maximises the limit of $\mathrm{slack}(\lambda)/\lambda$ as $\lambda \to 0$; for round 1, $E = e^{0.3864}$ and $m(0) = 1.505717$. The witness is a proposer's choice: any continuous M into $(0,1)$ that passes is a witness, and this one passes with the margins of Figure 1. Over round 1 the record has $M \le 0.3928$ and $F' \ge 0.756$.

Program 1. The file `tools/verify_gnnw_gai.py` (sha256 996c82c5500c…) is a single standard-library Python program: decimal intervals of 40 digits with every operation in a directed-rounding context and exp, log correctly rounded then widened one unit in the last place. On $[1/100, 1]$ it works piece by piece of m, bisecting adaptively; on an interval $[a, b]$ it encloses slack at the midpoint and bounds slack' over $[a, b]$ with the derivative written out (F'', M', $(\log X)'$, and $(\log Y)' = -\kappa(\log X)'$ on the branch of Lemma 1 in play), so that the mean-value form $\mathrm{slack}(\mathrm{mid}) \pm \frac{b-a}{2} \sup |\,\mathrm{slack}'\,|$ gives a certified lower bound; it also checks $F' > 0$, $M \in (0,1)$ and $\log X < 0$ on every interval. The branch parameter t of Y_f is located by a float solve and then bracketed by interval evaluation: the float is a proposer and the bracket is the authority. On $(0, 1/100]$ the three terms in $\lambda \log \lambda$ ($-\lambda \log \lambda$ in F, and $\frac{1}{2}\lambda \log \lambda$ each in $\lambda \log M$ and in $\lambda \log Y$ through $t = \lambda\tau$) are cancelled by hand, and $S = \mathrm{slack}\,/\lambda$ is enclosed on intervals that contain 0, with $\log(1+z)/z$ and $\log(1-z)/z$ bracketed by their series. An interval that cannot be decided is bisected; below a fixed width the program refuses, naming the interval.

Program 2. The file `instruments/gnnw/second.js` (sha256 9012c15a194f…) is a JavaScript program on dyadic big-float intervals of 120 bits with directed rounding and its own exp and log. It writes no derivative. On $[1/100, 1]$ it decides four monotonicity facts on each interval by enclosure — $F' > 0$, $F'' < 0$ and $M' > 0$ (so X decreases, every term of $(\log X)'$ being negative), and Y_f decreasing (Lemma 1) — and then uses only endpoint values:

$$\mathrm{slack}(\lambda) \;\ge\; F(a) + \tfrac{1}{2}\big[\log X(b) + b \log M(a) + b \log Y_f(X(a))\big] \qquad (\lambda \in [a, b]).$$

The branch parameter comes through $\beta(t) = f(t) - t f'(t)$ computed as written rather than through the rearranged form of program 1; the tail is handled as in program 1 with explicit remainders. The two programs share no line of code, no arithmetic layer and no bound; they share one author and one reading of [8].

The proposer. Rounds 2 to 6 were proposed by a floating-point optimiser (SLSQP; the code is `instruments/gnnw/propose/`): minimise $q(1)$ over polynomials q of degree 9 with $|c_i| \le 10$, subject to $\max_M \mathrm{slack}(\lambda) \ge \delta\lambda$ at 215 grid points and to the $\lambda \to 0$ limit, with $\delta = 5 \times 10^{-5}$, in the region of the round before. Its output rounded to six decimals, with the witness M built as above, is what the two programs decide. The optimiser is never an authority; its only role is to propose, and the record keeps its float sequences apart from the verdicts (Section 5.1).

Controls. The record is admitted only through a battery (27 checks, 7 forgeries that must be refused). Green controls: the paper's own $F_{0.03}$ passes Theorem 3 in its own region with this kind of M; the written derivative agrees with a central difference of the slack at eight points; the

tail form meets the direct slack $/\lambda$ at three points; $\log Y_f$ decreases across the three branches; the base $4e^{G_{\mathrm{AI}}(1)}$ computed directly agrees with $e^{F(1)}$ to 27 digits; the second program's record is for the code and the chain on disk and agrees. Red controls, each refused: the linear coefficient of G_{AI} moved from -0.3864 to -0.3870; a witness with $m(0) = 1.2$; the sixth coefficient lowered by 0.002; an M that reaches 1.2 at $\lambda = 1$; a float proposer that lies about a branch parameter; a round 2 whose last coefficient is lowered by 0.003; a non-concave polynomial offered as a region.

5 Results: the six rounds

Table 1 gives the polynomials; Table 2 the bases decided by each program; Table 3 the covers and the smallest certified margins. Every round is CERTIFIED by both programs. Program 1 decides the whole chain in 176 s (round 1 alone in 7.6 s) on 4,059 intervals; program 2, with its coarser endpoint bound, uses 181,977 intervals and 35 minutes. The region verdict recorded for every round is "strictly concave and increasing on (0, 1]; A < B". The final bases agree on 25 decimal digits: program 1 encloses c in $[3.772130762900366389246867 1139\ldots]$ and program 2 in $[3.77213076290036638924686 71,\ 3.77213076290036638924686 72]$.

round	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8	c_9
1	−0.3864	0.8347	−2.0156	2.7171	−1.7541	0.4522	—	—	—
2	−0.416545	0.646462	0.255568	−4.938579	10.000000	−7.007958	−1.007665	3.557723	−1.246039
3	−0.434002	0.835266	−0.514657	−3.672465	10.000000	−10.000000	3.360082	0.894786	−0.628130
4	−0.434013	0.830597	−0.499373	−3.688645	10.000000	−10.000000	3.384087	0.866940	−0.618928
5	−0.434220	0.831682	−0.501864	−3.686634	10.000000	−10.000000	3.382308	0.868797	−0.619493
6	−0.434241	0.831665	−0.501774	−3.686715	10.000000	−10.000000	3.382333	0.868789	−0.619496

Table 1: The polynomials $q_i(\lambda) = \sum_j c_j \lambda^j$ of the 6 rounds, $F_i = h + q_i e^{-\lambda}$. Round 1 is G_{AI} as printed in [8]; rounds 2–6 are the float proposer's, rounded to six decimals. The box constraint $|c_j| \le 10$ is active on 2 coefficients of the last round.

round	$\deg q$	$m(0)$	base $e^{F_i(1)}$, program 1	program 2, lower end of enclosure
1	6	1.505717	3.78232877553731	3.78232877553731162174280 04
2	9	1.496675	3.77547102153684	3.77547102153684571792204 46
3	9	1.491435	3.77257346178090	3.77257346178090806487108 55
4	9	1.491432	3.77227508535433	3.77227508535433496251839 54
5	9	1.491370	3.77215157829815	3.77215157829815423411940 64
6	9	1.491363	3.77213076290036	3.77213076290036638924686 71

Table 2: The bases, as recorded. Program 1's per-round bases are kept to 14 decimals in the record (its final enclosure is 28 digits wide, both ends equal to that length); program 2's enclosures are 25 digits, and the two ends of each differ only in the last digit. The bases decrease: $3.782328 \to 3.775471 \to 3.772573 \to 3.772275 \to 3.772151 \to 3.772130$.

The margins are thin and that is the content of the result. In round 1 the slack is about $6.31 \times 10^{-5}\,\lambda$ as $\lambda \to 0$ and $6.20 \times 10^{-5}\,\lambda$ near $\lambda = 0.93$ (Figure 1); the worst main interval of program 1 is $[0.6875, 0.69]$ with certified lower bound 9.55×10^{-7}. The later rounds are proposed at margin $\delta\lambda$ with $\delta = 5 \times 10^{-5}$ and decided at the certified lower bounds of Table 3.

	program 1				program 2			
round	tail	main	$\min S$	$\min$ slack	tail	main	$\min S$	$\min$ slack
1	79	336	4.7×10^{-6}	9.6×10^{-7}	79	11,322	4.7×10^{-6}	3.7×10^{-8}
2	10	693	3.6×10^{-4}	1.4×10^{-7}	10	39,820	3.0×10^{-4}	2.3×10^{-10}
3	16	736	4.9×10^{-4}	6.6×10^{-8}	16	34,758	4.6×10^{-4}	1.3×10^{-11}
4	12	718	7.1×10^{-4}	1.0×10^{-7}	12	31,982	6.7×10^{-4}	5.4×10^{-10}
5	12	717	4.7×10^{-4}	1.7×10^{-7}	12	31,713	4.3×10^{-4}	2.2×10^{-10}
6	12	718	5.3×10^{-4}	1.0×10^{-8}	12	32,241	4.9×10^{-4}	3.8×10^{-11}

Table 3: The covers. "tail" counts intervals of $(0, 1/100]$, on which $S = \text{slack}/\lambda$ is decided; "main" the intervals of $[1/100, 1]$. The columns $\min S$ and $\min$ slack are the smallest certified lower bounds over the cover (program 1: the mean-value lower bound; program 2: the endpoint bound), not the minima of the functions.

5.1 The float sequences, and where the iteration goes

The proposer, run on its own and never decided, iterates past the certified rounds. Table 4 and Figure 2 record its sequences at degree six (5 iterations) and degree nine (8 iterations), each round in the float region of the round before, with the same margin δ. Degree six settles near 3.7732 (last step 4.3×10^{-5}); degree nine near 3.77213 (last step 2.6×10^{-7}), and the certified round 6 lies 4.6×10^{-6} above the last float iterate. These are observations in floating point about the shape of the iteration, not bounds: they say that further rounds of this kind, in this parametrisation of F, would move the base by less than 10^{-5}. The authors of [8] expect that going below 3.75 needs new ideas; nothing here disagrees.

iteration	degree 6 (float, not decided)	degree 9 (float, not decided)
0	3.782329	3.7823288
1	3.775910	3.7754705
2	3.773963	3.7725741
3	3.773382	3.7722750
4	3.773223	3.7721512
5	3.773180	3.7721329
6	—	3.7721274
7	—	3.7721264
8	—	3.7721261

Table 4: The float proposer's bases, iteration 0 being G_{AI}. Rounds 2–6 of Theorem 1 are iterations 1–5 of the degree-9 column, rounded to six decimals and decided.

6 The HorizonMath certificate

6.1 The problem and the certificate

HorizonMath [9] (version 2603.15617v2, pinned by sha256 9616325eadb6…; its code at commit 4ef0b61a of `github.com/ewang26/HorizonMath`) states the problem in the framework of Theorem 3, with $F(\lambda) = h(\lambda) + p(\lambda)e^{-\lambda}$ for a polynomial p without constant term and $c = e^{F(1)}$,

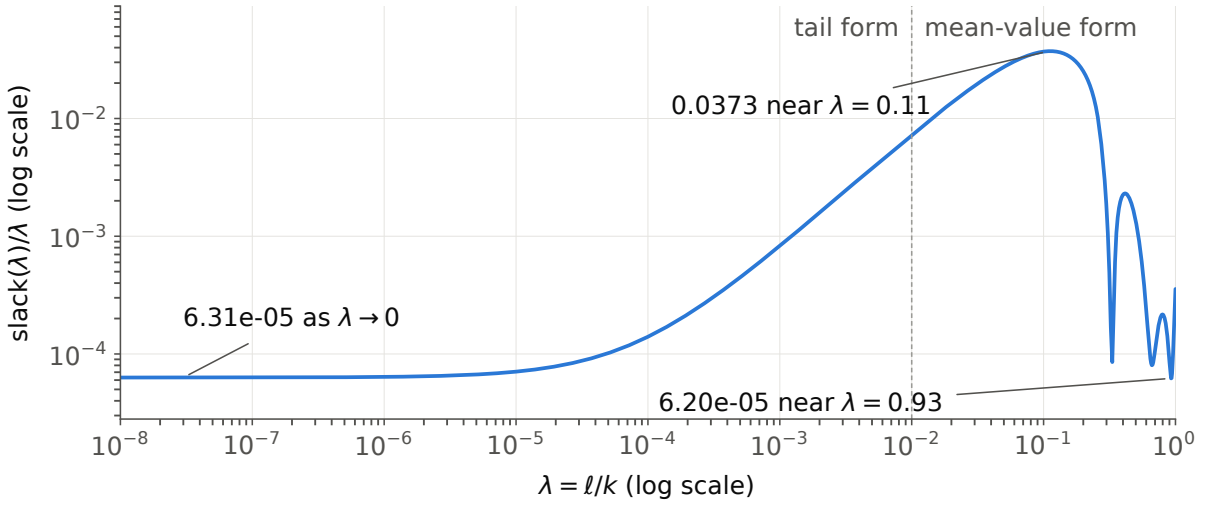


Figure 1: Round 1: the slack of Theorem 3's inequality divided by λ, as the lower end of an interval enclosure at each of 160 points (the decision is the cover of Table 3, not these points). It is flat near 6.31×10^{-5} for small λ, rises to 0.0373 near $\lambda = 0.11$, falls to 6.20×10^{-5} near $\lambda = 0.93$, and ends at 3.56×10^{-4} at $\lambda = 1$. The flat left end and the three dips between 0.6 and 0.95 are where a coarser witness, or a slightly more ambitious q, is refused; the control that moves the linear coefficient to -0.3870 fails at the left end.

and verifies the condition $(X, Y) \in R$ through an inner approximation: a rate function

$$U(\mu) = h(\mu) + \big(-0.25\mu + 0.033\mu^2 + 0.08\mu^3\big)e^{-\mu},$$

a set of pairs with $-\log x - \mu \log y \geq U(\mu)$ for all $\mu \in (0, 1]$ — the problem calls it R_0; we write R_0' to keep it apart from the R_0 of (1) — and the rule, in the problem's words, "Since $R(k, \ell) = R(\ell, k)$, the pair (x, y) is accepted if either $(x, y) \in R_0$ or $(y, x) \in R_0$." The function U is $F_{0.033}$, the third stage of the iteration in the first version of [8]; it is a weaker bound than Theorem 1's $F_{0.03}$ and is implied by it, so the problem's inner approximation rests on a proved bound.

The certificate credited to GPT-5.4 Pro, reproduced by GPT-5.6 Sol Max is printed as code (Appendix A.3 of [9]) and was rebuilt here from that recipe. Its polynomial is

$$p(\lambda) = \sum_{j=1}^{5} c_j \lambda^j, \qquad (c_1, \ldots, c_5) = (-0.25,\ 0.033,\ 0.08,\ 0,\ -0.0778),$$

the cubic of U plus $-0.0778\,\lambda^5$; M and Y are constant on 200 intervals of $(10^{-3}, 1]$, with Y set on each interval to $(1 - 0.0012)\min\{1,\ e^{-U(1)}/X\}$ at the interval's left end, in the certificate's own words "0.12% below the active $xy = e^{-U(1)}$ branch". The benchmark's checker accepts it and reports $c = e^{F(1)} = 3.69608391263$ (its whole output: "Valid split certificate; c = e^{F(1)} = 3.69608391263; polynomial_coeffs degree = 5; analytic tail endpoint = 0.00010846; worst small-lambda slack = 0.000272305; worst large R_0 slack = 0.000951234; worst large main slack = 0.000318787"); the paper prints $c = 3.6960839126\ldots$; our enclosure of $e^{F(1)}$ reproduces the digits, $[3.6960839126\ldots]$.

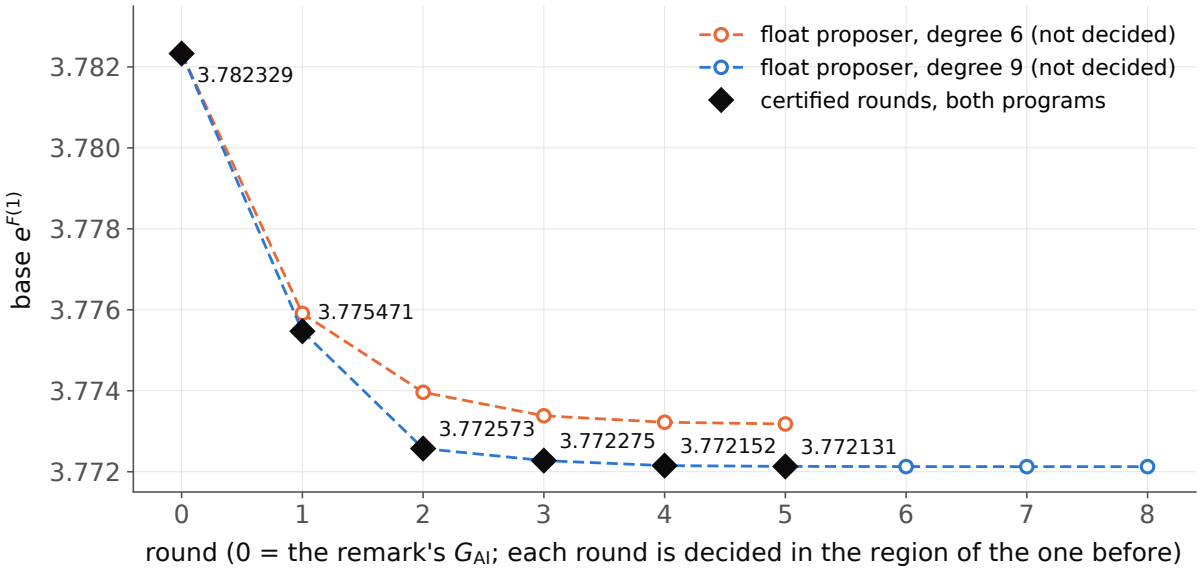


Figure 2: The certified bases of the 6 rounds (filled diamonds, both programs) beside the float proposer's sequences (hollow markers on dashed lines, never decided). Decided and proposed values share neither colour nor mark.

6.2 Two inequalities, and a rule that asks for one

A bound $R(k,\ell) \le e^{U(\ell/k)k+o(k)}$ is known for $\ell \le k$. By Observation 10(4) of [8] it places (x,y) in R when $R(k,\ell) \le x^{-k-o(k)}y^{-\ell-o(\ell)}$ for *all* k,ℓ, that is when, for every $s \in (0,1]$,

$$-\log x - s\log y \;\ge\; U(s) \qquad \text{(the pairs with } \ell \le k\text{)}, \tag{2}$$

$$-\log y - s\log x \;\ge\; U(s) \qquad \text{(the pairs with } \ell > k\text{, read through } R(k,\ell) = R(\ell,k)\text{)}. \tag{3}$$

Lemma 1 proves both before it places a point. The problem's R_0' is (2) alone, and its rule accepts (x,y) when (2) holds *or* (3) holds. Symmetry of $R(k,\ell)$ makes R symmetric; it does not make a one-sided test two-sided. The consequence is concrete. U increases on $(0,1]$ (as in (35) of [8], $U' \ge \log 2 - \frac{1}{4}$), so (2) holds for every y as soon as $-\log x \ge U(1)$, i.e. $x \le e^{-U(1)} = 0.26292279\ldots$ — the certificate's own constant $c_{\mathrm{hyp}} = 0.26292278641656774$. The rule therefore accepts the whole strip $\{x \le 0.26292\} \times (0,1)$, for instance $(0.26, 0.9999)$, which R cannot contain (Lemma 2 below). The certificate places its pairs on that strip wherever X is small, and $X(1) = 0.22745$ is.

6.3 The exclusion

Lemma 2 (Erdős [3], first moment)**.** *For $k,m \ge 3$ and $0 < p < 1$ let $N = \lfloor \min(p^{-(k-1)/2}, (1-p)^{-(m-1)/2}) \rfloor$. Then $R(k,m) > N$. Consequently, if $(x,y) \in R$, then for every $e > 0$ and $0 < p < 1$,*

$$e\,(-\log x) + (-\log y) \;\ge\; \min\!\big(\tfrac{e}{2}(-\log p),\ \tfrac{1}{2}(-\log(1-p))\big),$$

and the same with x and y exchanged.

Proof. Colour K_N red with probability p independently. The expected number of red K_k is at most $N^k p^{\binom{k}{2}}/k! \le 1/6$ and of blue K_m at most $1/6$, so some colouring has neither. For the

consequence, let $(x, y) \in R$ and $d > 0$; since R is the closure of R_0 and R_0 is closed downward, $(x - d, y - d) \in R_0$, so $N < R(k, m) \le (x - d)^{-k}(y - d)^{-m}$ for all large $k + m$. Take $k = \lceil em \rceil$, divide the logarithms by m, let $m \to \infty$ and then $d \to 0$. The exchanged form is the regime $m = \lceil ek \rceil$. □

The decision is one rational pair (e, p) per point: the lower end of an enclosure of the right side exceeds the upper end of an enclosure of the left side, in decimal intervals of 60 digits with every rounding outward (`instruments/horizonmath/ramsey_region.py`, written from the two papers and sharing no code with the benchmark's validator). A float search proposes (e, p); the interval comparison is the authority. At $\lambda = 1$, with $e = 13/2000$ and $p = 11981/500000$ in the regime $k = \lceil em \rceil$, Erdős's rate is at least 0.012127 per m while the pair allows at most 0.010826: a gap of 0.00130. The pair is not in R; the second hypothesis of Theorem 3 fails at the point where $c = e^{F(1)}$ is read; Proposition 1 follows.

Across the certificate (Figure 3, Table 5): 56 of 201 points are outside R, each with its own (e, p) — among them the first interval, at $\lambda = 0.00102$, where $(X, Y) = (0.99769, 0.26321)$ is excluded with $e = 19/2000$, $p = 8033/250000$; 128 more are shown by one rational s at which $U(s)$ exceeds the line (2) or (3) not to be placed in R by the problem's own U (they may or may not lie in R by some other bound); and 17 are not refuted here, the points with $\lambda \in [0.236, 0.374]$ and $X \in [0.4365, 0.5988]$, near the window $[0.4657, 0.5645]$ where the hyperbola branch of Lemma 1 is legitimate for $f = U$ (that window is computed in floating point for the figure and is not part of any verdict).

λ	$X(\lambda)$	$Y(\lambda)$	decided	witness
0.0010	0.99769	0.26321	outside R	$e = 19/2000$, $p = 8033/250000$: gap 0.00134
0.0058	0.98706	0.26601	not placed by U	$s = 319/1000$: $U(s)$ exceeds the line by 0.2406
0.0331	0.92892	0.28245	not placed by U	$s = 703/2000$: $U(s)$ exceeds the line by 0.2000
0.1407	0.73107	0.35746	not placed by U	$s = 541/1000$: $U(s)$ exceeds the line by 0.0632
0.3134	0.50087	0.52209	not refuted here	—
0.5725	0.26328	0.99325	not placed by U	$s = 157/500$: $U(s)$ exceeds the line by 0.2436
0.8316	0.26070	0.99880	outside R	$e = 23/2500$, $p = 31483/1000000$: gap 0.00234
1.0000	0.22745	0.99880	outside R	$e = 13/2000$, $p = 11981/500000$: gap 0.00130

Table 5: Eight of the 201 decided points, X as the lower end of its enclosure. A witness (e, p) excludes the pair from R by Lemma 2; a witness s shows the problem's U does not place it.

6.4 The checker, as published and with one token changed

As an observation, recorded in the corpus and not part of any verdict: the benchmark's validator (`validators/ramsey_asymptotic.py` at the pinned commit), run as published on 2026-09-29, accepts the rebuilt certificate in 107 s. Its line 299 reads `return max(mp.mpf(0), min(bu, bs))`, the better of the two orientations' margins; with `max(bu, bs)` in its place, so that both orientations must clear their thresholds as membership in R requires, it refuses the certificate in 0.2 s, on its first interval $[0.001, 0.0010014]$. The defect is in the problem statement, which gives the either-orientation rule in words, and in its checker; the model optimised against the problem as stated, and the certificate reads as a faithful solution of that problem. The verification that relied on the checker is what the record contradicts.

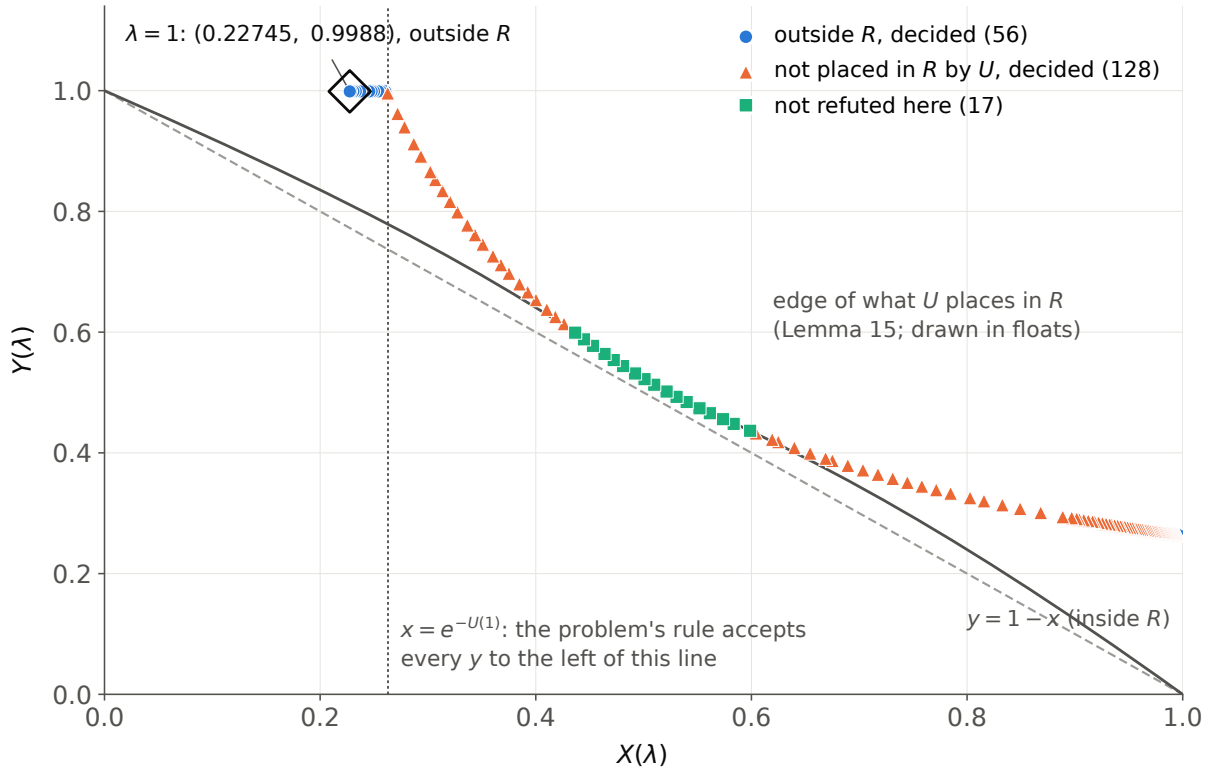


Figure 3: The certificate's 201 pairs $(X(\lambda), Y(\lambda))$, each at the midpoint of an interval on which M and Y are constant, and $\lambda = 1$ itself (the diamond). Marks and colours repeat the three verdicts, which are decided; the solid curve is the edge of what U places in R by Lemma 1, drawn in floats and not decided; the dashed diagonal $y = 1 - x$ is inside R by the Erdős–Szekeres bound. Left of the dotted line $x = e^{-U(1)}$ the problem's rule accepts every y.

Controls. The battery for this ledger (22 checks, 7 red controls, four of them on this row) requires that the 99 Erdős–Szekeres pairs $(j/100, 1 - j/100)$ and the forty points $(X_0, 1 - X_0)$ of Lemma 16(1) of [8] are never excluded from R nor left unplaced by U; that $(e^{-U(1)} - 10^{-6}, 0.9999)$, which the rule accepts, is excluded; that the certificate's pair at $\lambda = 1$ is excluded; and that a float proposer claiming $(\frac{1}{2}, \frac{1}{2})$ outside R, or unplaced by U, is overruled by the interval check. The same ledger decides the benchmark's other printed construction, a 128-triangle Kakeya set, whose union area is exactly $6008623/55050240 = 0.1091479891\ldots$ (CERTIFIED), and records the three closed forms it credits to a second model as NEEDS DATA because they are not printed; neither is in this paper's scope.

7 What is not claimed

- Theorem 14, Lemma 15 and Theorem 1 of [8] are used, not re-proved, and the $o(k)$ in Theorem 1 is theirs, including its uniformity in $\ell \le k$ on which the induction through Lemma 15 depends. Round i rests on the bound round $i - 1$ establishes; the chain is as strong as its first link, which is the paper's Theorem 1.
- Nothing here is the best known bound on $R(k, k)$. Theorem 1 continues the iteration of [8];

$R(k,k) \le 3.769^{\,k+o(k)}$ [10] and $R(k,k) \le 3.69507^{\,k}$ [11] are lower, and neither is checked here.

- No inequality about Ramsey numbers is refuted. Proposition 1 refutes a certificate: whether $R(k,k) \le (3.6960\ldots)^{k+o(k)}$, or any base below 3.7992, holds is untouched by it (the bound of [11], not checked here, would imply the former). Whether some certificate of the benchmark's quintic form, with pairs inside R, beats 3.7992 is open; the Erdős bound alone does not exclude the quintic F of Section 6 for every M.
- Nothing is claimed about optimality. The float observation of Section 5.1 concerns this parametrisation of F and this proposer; a different family of F, or a different M, is not excluded by it.
- The two programs are independent in language, arithmetic and bounding method, but share an author and a reading of the paper. A check by the authors of [8], or by anyone running the one-file verifier, is the next step; nothing has been sent to the authors of either paper.
- This document is machine-derived and not peer-reviewed. Every number in it is interpolated from the records named below; the author asserts nothing numerical beyond what they hold.

Points for review. (i) The reading of Theorem 14's hypothesis "$F : (0,1] \to \mathbb{R}_+$ smooth" as satisfied by $h+qe^{-\lambda}$ with positivity implied by the strict inequality. (ii) Whether the hypotheses of Lemma 15 decided for each F_{i-1} (Section 3) are all of them, and whether applying the lemma to a bound obtained from Theorem 14 (uniform $o(k)$) is exactly its hypothesis (29). (iii) The reading of the remark as the statement that Theorem 14's hypotheses can be met for $F = h+G_{\mathrm{AI}}$ with Y from Remark 17 — the remark names no M or Y, so a different reading would decide a different claim. (iv) The reading of R as two-sided, on which Proposition 1 turns; it is the definition (12) of [8] and the practice of its Lemma 15. (v) The record's reproduction line names the verifier as `verify/verify_gnnw_gai.py`; the file is at `tools/verify_gnnw_gai.py`, and the sha256 above is of that file.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records:

- `certs/gnnw-certificate.json` — round 1, generated 2026-09-29;
- `certs/gnnw-chain-certificate.json` — the 6 rounds, both programs, generated 2026-09-29 (the second program's run of 2026-09-29 is `corpus/gnnw/second-run.json`, pinned by the sha256 of its code and of the chain);
- `certs/horizonmath-ledger.json` — the benchmark's certificate, generated 2026-09-29.

The corpus is `corpus/gnnw` — the paper arXiv 2407.19026v2 pinned by sha256 2d5be61d0442… and its version 1 by 148e722db641…, neither stored; the remark transcribed; the witness nodes; the chain, sha256 4912112b1b17… — and `corpus/horizonmath` — the paper and the benchmark's code pinned, not stored; the printed recipe transcribed and the certificate rebuilt from it by `rebuild_ramsey_a3.py`. From the repository root (commit 16849460e608):

- round 1 with the standard library alone, about 7.6 s:
  `python3 tools/verify_gnnw_gai.py certs/gnnw-certificate.json`
- the chain, about 176 s:
  `python3 tools/verify_gnnw_gai.py certs/gnnw-chain-certificate.json`
- the second program, about 35 minutes:
  `node instruments/gnnw/second.js corpus/gnnw/chain.json`
- re-derive both records: `python3 tools/run-gnnw-ledger.py --check`
- the 27 checks and 7 refused forgeries: `python3 instruments/gnnw/battery.py`
- the benchmark's ledger re-derived: `python3 tools/run-horizonmath-ledger.py --check`
- its 22 checks: `python3 instruments/horizonmath/battery.py`
- the proposer (numpy and scipy); its second file reproduces the chain byte for byte:
  `instruments/gnnw/propose/iterate2.py`, `instruments/gnnw/propose/buildchain.py`
- the macros and the figures of this document, from the records, and the document itself:
  `node tools/paper-numbers/ramsey-chain.js`
  `instruments/hseva/.venv/bin/python tools/paper-numbers/ramsey-chain-figs.py`
  `node tools/build-paper-tex.js ramsey-chain`

Pages built from the same records: `https://carlostoledo.co/reports/diagonal-ramsey.html` and `https://carlostoledo.co/reports/horizonmath.html`.

References

[1] F. P. Ramsey, On a problem of formal logic, Proc. London Math. Soc. (2) 30 (1930), 264–286.

[2] P. Erdős, G. Szekeres, A combinatorial problem in geometry, Compositio Math. 2 (1935), 463–470.

[3] P. Erdős, Some remarks on the theory of graphs, Bull. Amer. Math. Soc. 53 (1947), 292–294.

[4] A. Thomason, An upper bound for some Ramsey numbers, J. Graph Theory 12 (1988), 509–517.

[5] D. Conlon, A new upper bound for diagonal Ramsey numbers, Ann. of Math. (2) 170 (2009), 941–960.

[6] A. Sah, Diagonal Ramsey via effective quasirandomness, Duke Math. J. 172 (2023).

[7] M. Campos, S. Griffiths, R. Morris, J. Sahasrabudhe, An exponential improvement for diagonal Ramsey, arXiv:2303.09521 (2023).

[8] P. Gupta, N. Ndiaye, S. Norin, L. Wei, Optimizing the CGMS upper bound on Ramsey numbers, arXiv:2407.19026; version 2 of 29 August 2026, pinned by sha256 2d5be61d0442… (`corpus/gnnw/meta.json`); its main results are stated to be formalised in Lean 4 at `https://github.com/snorin239/RamseyLean`.

[9] E. Y. Wang, S. R. Motwani, J. V. Roggeveen, E. Hodges, D. Jayalath, C. London, K. Ramakrishnan, J. Foerster, C. Zhang, F. Cipcigan, P. Torr, A. Abate, Measuring AI progress toward mathematical discovery with automatic verification (HorizonMath), arXiv:2603.15617; version 2 of 10 September 2026, pinned by sha256 9616325eadb6…; code at `https://github.com/ewang26/HorizonMath`, commit 4ef0b61a (`corpus/horizonmath/meta.json`).

[10] C. Neisler, R. M. J. Shin, S. Sukhatankar, A self-consistent bootstrap for the book algorithm and the bound $R(k,k) \leq 3.769^{k+o(k)}$, revision v3, 2 September 2026, with a Lean 4 formalization at `https://github.com/wamlat/RamseyLean-bootstrap`, archived at doi:10.5281/zenodo.22263823 (3 September 2026); README, Zenodo record and the paper source's sha256 pinned in `corpus/sources/ramsey`.

[11] Z. Lu, S. Wang, Retained-Set Descent for Diagonal Ramsey Numbers, arXiv:2609.14525 (13 September 2026); the abstract page pinned in `corpus/sources/ramsey`.

[12] R. E. Moore, Interval Analysis, Prentice-Hall, 1966.

[13] W. Tucker, Validated Numerics: A Short Introduction to Rigorous Computations, Princeton University Press, 2011.